

A pure transverse-momentum-space prescription for high-accuracy TMD extractions

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Abstract

We formulate a transverse-momentum-space representation of Collins/CSS2 transverse-momentum-dependent (TMD) factorization suitable for high-accuracy extractions without numerical transformations to impact-parameter space. The construction, denoted $\mathcal{K}_{\text{CSS2}}$, uses two-dimensional plus distributions and transverse convolution kernels as the working objects. The local Sudakov factor in b_T space is replaced by a convolution-exponential Green function in k_T space, while the TMD matching coefficients are imported by a coefficient-level map $L_b^m \mapsto \mathfrak{L}_{m-1}(\mathbf{k}; \mu)$. We define the perturbative accuracy label $N^k\text{LL}'$ by ingredient equivalence to the parent CSS2 calculation: the same cusp and noncusp anomalous dimensions, Collins-Soper kernel, hard factors, TMD matching coefficients, DGLAP kernels, and fixed-order subtractions must be included, but represented in k_T space. The nonperturbative input is left as a model-agnostic module; neural networks are one possible parametrization, not part of the perturbative accuracy definition. We also give the generic process-level W operator, the $W + Y$ matching prescription, regulated normalization and moment constraints, and a validation hierarchy for future numerical implementations. The result is a formal prescription for pure- k_T TMD extractions capable of carrying a $N^3\text{LL}'$ perturbative label once the required physical three-loop coefficient payloads are supplied and validated.

I. INTRODUCTION

TMD factorization organizes the small-transverse-momentum region of color-singlet production and semi-inclusive reactions in terms of hard factors, TMD parton distributions or fragmentation functions, Collins-Soper evolution, and matching onto collinear parton distributions at short transverse distances [1–4]. The standard high-accuracy implementation is formulated in impact-parameter space, where the transverse convolution becomes a product and the Sudakov factor is local in b_T . This feature is technically powerful, and modern phenomenology has reached high logarithmic accuracy in b_T -space formulations [5–7].

There are nevertheless good reasons to develop an equivalent formulation that remains directly in transverse momentum. First, momentum-space representations interface naturally with measured q_T bins and with flexible nonperturbative models in k_T . Second, a pure- k_T implementation avoids fitting an object in one representation and repeatedly transforming it to another. Third, recent neural-network and inverse-problem methods have shown that direct

momentum-space reconstruction can be practical, although not yet with the perturbative content required for an $N^k\text{LL}$ label [8, 9]. Finally, direct momentum-space and distribution-space resummation methods already demonstrate that logarithmic resummation need not be tied to numerical b_T integration [10–14].

The purpose of this paper is to define a rigorous pure- k_T prescription, not to report a fitted extraction. We construct a k_T -space representation of the Collins/CSS2 scheme, denoted $\mathcal{K}_{\text{CSS2}}$, and show how the standard perturbative ingredients are represented by distribution-valued kernels and transverse convolution exponentials. The central claim is deliberately limited:

A calculation in $\mathcal{K}_{\text{CSS2}}$ has the same $N^k\text{LL}'$ perturbative accuracy as the corresponding CSS2 calculation when, and only when, the same perturbative ingredients are included and the fixed-order expansion is validated in k_T space.

Thus, the paper supplies the formal prescription needed for a future $N^3\text{LL}'$ extraction. A numerical extraction at that accuracy additionally requires staged physical three-loop matching coefficients, code-level validation, data treatment, and a nonperturbative fit.

II. DEFINITION OF THE $\mathcal{K}_{\text{CSS2}}$ REPRESENTATION

We define $\mathcal{K}_{\text{CSS2}}$ as the direct transverse-momentum-space representation of the CSS2 TMD scheme. It is not introduced as a new operator scheme with new finite subtractions. Instead,

$$F_{i/h}^{\mathcal{K}_{\text{CSS2}}}(x, \mathbf{k}; \mu, \zeta) \equiv \int \frac{d^2\mathbf{b}}{(2\pi)^2} e^{-i\mathbf{b}\cdot\mathbf{k}} \tilde{F}_{i/h}^{\text{CSS2}}(x, \mathbf{b}; \mu, \zeta), \quad (1)$$

with analogous definitions for the Collins-Soper kernel and matching coefficients,

$$\mathcal{D}_i^{\mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu) \equiv \mathcal{F}_{b \rightarrow k}^{-1} \left[\tilde{D}_i^{\text{CSS2}}(b_T; \mu) \right], \quad (2)$$

$$\mathcal{C}_{i \leftarrow j}^{\mathcal{K}_{\text{CSS2}}}(z, \mathbf{k}; \mu, \zeta) \equiv \mathcal{F}_{b \rightarrow k}^{-1} \left[\tilde{C}_{i \leftarrow j}^{\text{CSS2}}(z, b_T; \mu, \zeta) \right]. \quad (3)$$

Equation (1) is a definition of the analytic distribution dictionary. It is not a prescription to perform numerical Fourier-Bessel transforms during the fit. Once the coefficient functions and evolution kernels are projected into the distribution basis, all subsequent operations are transverse convolutions in k_T .

The CSS2 W term for a Drell-Yan-like process,

$$W^{\text{CSS2}}(\mathbf{q}) = \sum_{ij} H_{ij}(Q, \mu) \int \frac{d^2 \mathbf{b}}{(2\pi)^2} e^{-i\mathbf{b} \cdot \mathbf{q}} \tilde{F}_{i/A}(x_A, \mathbf{b}; \mu, \zeta_A) \tilde{F}_{j/B}(x_B, \mathbf{b}; \mu, \zeta_B), \quad (4)$$

becomes

$$W^{\mathcal{K}_{\text{CSS2}}}(\mathbf{q}) = \sum_{ij} H_{ij}(Q, \mu) \left[F_{i/A}^{\mathcal{K}_{\text{CSS2}}} \otimes_T F_{j/B}^{\mathcal{K}_{\text{CSS2}}} \right](\mathbf{q}), \quad (5)$$

where

$$[A \otimes_T B](\mathbf{q}) = \int d^2 \mathbf{k} A(\mathbf{k}) B(\mathbf{q} - \mathbf{k}). \quad (6)$$

The hard factor is the parent CSS2 hard factor. If a genuinely new k_T -adapted child scheme is later introduced, it must be related to $\mathcal{K}_{\text{CSS2}}$ by explicit finite transverse convolution kernels, not by an implicit reshuffling of terms.

III. DISTRIBUTION BASIS AND CONVOLUTION ALGEBRA

The natural k_T -space objects are distributions. We use the logarithmic two-dimensional plus distributions $\mathcal{L}_n(\mathbf{k}; \mu)$, together with a derivative basis $\mathfrak{L}_n$ defined by its Fourier image,

$$\mathcal{F}_{k \rightarrow b}[\mathfrak{L}_n(\mathbf{k}; \mu)] = L_b^{n+1}, \quad L_b \equiv \ln \frac{\mu^2 b_T^2}{b_0^2}, \quad b_0 = 2e^{-\gamma_E}. \quad (7)$$

The first relation to the conventional basis is

$$\mathfrak{L}_0 = -\mathcal{L}_0, \quad (8)$$

with higher-order relations containing the corresponding zeta-valued contact terms. The derivative basis has the simple convolution rule

$$\boxed{\mathfrak{L}_m \otimes_T \mathfrak{L}_n = \mathfrak{L}_{m+n+1}.} \quad (9)$$

Equivalently, if $\mathcal{G}_\eta$ is the analytic generator satisfying

$$\mathcal{F}_{k \rightarrow b}[\mathcal{G}_\eta] = e^{\eta L_b}, \quad (10)$$

then

$$\mathcal{G}_\eta \otimes_T \mathcal{G}_\xi = \mathcal{G}_{\eta+\xi}, \quad \text{Exp}_{\otimes_T} \{\eta \mathfrak{L}_0\} = \mathcal{G}_\eta. \quad (11)$$

More generally,

$$\mathcal{F}_{k \rightarrow b} \left[\text{Exp}_{\otimes_T} \left\{ \sum_{n \geq 0} r_n \mathfrak{L}_n \right\} \right] = \exp \left[\sum_{n \geq 0} r_n L_b^{n+1} \right]. \quad (12)$$

This identity is the core mechanism by which the local b_T -space Sudakov exponent is represented in k_T space without numerical b_T transformations.

For comparisons with data, distributions should be evaluated on test functions or bins, not pointwise at the origin. For an annular bin $q_a < |\mathbf{q}| < q_b$, define

$$\mathcal{B}_n(q_a, q_b; Q) = \int_{q_a < |\mathbf{q}| < q_b} d^2\mathbf{q} \mathcal{L}_n(\mathbf{q}; Q). \quad (13)$$

For $q_a > 0$ this reduces to the ordinary integral over the real-emission function, while the bin touching the origin contains the plus-prescription subtraction. This bin-level treatment is essential for numerical stability and for comparison with experimental data.

IV. EVOLUTION IN k_T SPACE

The CSS2 Collins-Soper equation becomes nonlocal in k_T space,

$$\frac{\partial}{\partial \ln \sqrt{\zeta}} F_{i/h}(x, \mathbf{k}; \mu, \zeta) = -\mathcal{D}_i(\mathbf{k}; \mu) \otimes_T F_{i/h}(x, \mathbf{k}; \mu, \zeta), \quad (14)$$

while the μ evolution remains multiplicative in transverse momentum,

$$\frac{\partial}{\partial \ln \mu} F_{i/h}(x, \mathbf{k}; \mu, \zeta) = \gamma_F^i(\mu, \zeta) F_{i/h}(x, \mathbf{k}; \mu, \zeta). \quad (15)$$

At a final scale (μ_f, ζ_f) and initial scale (μ_i, ζ_i) , a convenient endpoint representation of the running-coupling Green function is

$$\mathcal{U}_i(\mathbf{k}; f, i) = \exp[R_i(\mu_f, \mu_i; \zeta_i) - \rho D_i^\delta(a_f)] \text{Exp}_{\otimes_T} \left\{ -\rho \sum_{n \geq 0} D_{i,n}(a_f) \mathcal{L}_n(\mathbf{k}; \mu_f) \right\}, \quad (16)$$

where

$$\rho = \ln \sqrt{\frac{\zeta_f}{\zeta_i}}, \quad a_f = \frac{\alpha_s(\mu_f)}{4\pi}. \quad (17)$$

The scalar exponent is written in terms of the standard evolution integrals

$$R_i = A_{\gamma_i} - \ell_i A_{\Gamma_i} + 2S_{\Gamma_i}, \quad \ell_i = \ln \frac{\zeta_i}{\mu_i^2}. \quad (18)$$

The coefficient functions $D_{i,n}(a_f)$ are determined by the perturbative Collins-Soper kernel in the derivative basis,

$$\mathcal{D}_i(\mathbf{k}; \mu) = D_i^\delta(a_\mu) \delta^{(2)}(\mathbf{k}) + \sum_{n \geq 0} D_{i,n}(a_\mu) \mathcal{L}_n(\mathbf{k}; \mu). \quad (19)$$

Through two loops,

$$\mathcal{D}_i(\mathbf{k}; \mu) = a_s \Gamma_0^i \mathfrak{L}_0 + a_s^2 \left[\frac{1}{2} \beta_0 \Gamma_0^i \mathfrak{L}_1 + \Gamma_1^i \mathfrak{L}_0 + d_i^{(2)} \delta^{(2)} \right] + \mathcal{O}(a_s^3). \quad (20)$$

The three-loop structure is fixed by the three-loop rapidity anomalous dimension together with the RG equation for $\mathcal{D}_i$; its explicit finite contact term is an imported perturbative ingredient at $N^3\text{LL}'$.

V. MATCHING AND THE ACCURACY LABEL

At perturbative transverse momentum, the TMD matches onto collinear PDFs,

$$F_{i/h}^{\text{pert}}(x, \mathbf{k}; \mu, \zeta) = \sum_j \mathcal{C}_{i \leftarrow j}^{\mathcal{K}_{\text{CSS2}}}(\mathbf{k}; \mu, \zeta) \otimes_x f_{j/h}(\mu). \quad (21)$$

The coefficient-level import rule is mechanical. If the parent CSS2 coefficient is supplied as

$$\tilde{C}_{i \leftarrow j}^{(n), \text{CSS2}}(z, L_b, \ell_\zeta) = \sum_{m=0}^{2n} \tilde{c}_{i \leftarrow j}^{(n,m)}(z; \ell_\zeta) L_b^m, \quad \ell_\zeta = \ln \frac{\zeta}{\mu^2}, \quad (22)$$

then

$$\boxed{\mathcal{C}_{i \leftarrow j}^{(n), \mathcal{K}_{\text{CSS2}}} = \tilde{c}_{i \leftarrow j}^{(n,0)} \delta^{(2)} + \sum_{m=1}^{2n} \tilde{c}_{i \leftarrow j}^{(n,m)} \mathfrak{L}_{m-1}.} \quad (23)$$

No numerical Fourier-Bessel transform appears in Eq. (23). At $N^3\text{LO}$ the highest logarithmic power is L_b^6 , so the k_T basis must support $\mathfrak{L}_5$.

We define the $N^k\text{LL}'$ label in $\mathcal{K}_{\text{CSS2}}$ by ingredient equivalence to the parent CSS2 calculation. In the convention used here,

$$N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}} = N^3\text{LL evolution} + \text{three-loop hard and TMD matching boundary functions}. \quad (24)$$

The required ingredients are summarized in Table I. The prime is not a cosmetic label: it requires the fixed-order boundary functions at the same order as the claimed primed accuracy.

High-order ingredients entering this manifest are available in the literature: three-loop TMD matching coefficients for unpolarized TMDPDFs and TMDFFs [15–17], rapidity anomalous dimensions through three loops and beyond [18–20], three-loop splitting functions [21, 22], and high-order hard functions from quark and gluon form factors [23]. This paper does not print the long three-loop coefficient tables; they should be imported from source-stamped ancillary files and converted with Eq. (23).

TABLE I. Perturbative ingredient manifest for a $N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ singular W operator in the convention used in this paper. Equivalent conventions are possible, but the k_T implementation must match the chosen parent CSS2 convention ingredient by ingredient.

Ingredient	Required order for $N^3\text{LL}'$
Cusp anomalous dimension Γ_{cusp}	four loops
QCD beta function	four loops
TMD noncusp anomalous dimension γ_F	three loops
Collins-Soper kernel / rapidity anomalous dimension	three loops
Hard factor H	$\mathcal{O}(a_s^3)$
TMD matching coefficients $\mathcal{C}_{i \leftarrow j}$	$\mathcal{O}(a_s^3)$
DGLAP kernels in boundary evolution	$P^{(0)}, P^{(1)}, P^{(2)}$
W -term fixed-order expansion gate	through $\mathcal{O}(a_s^3)$
Y term for full-spectrum prediction	matching-consistent fixed order

We distinguish a formal prescription from a numerical implementation. A formal $N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ prescription is specified by Eqs. (12), (16), and (23) plus the ingredient manifest. A numerical implementation may claim $N^3\text{LL}'$ only after the physical three-loop payload is staged, converted, hashed, and validated by expansion through $\mathcal{O}(a_s^3)$.

VI. PROCESS-LEVEL W OPERATOR AND FULL-SPECTRUM MATCHING

The process-level singular operator is

$$W_{AB \rightarrow F}^{\mathcal{K}_{\text{CSS2}}}(\mathbf{q}; Q) = \sum_{ij} H_{ij \rightarrow F}(Q, \mu) \left[\mathbb{F}_{i/A}^{\mathcal{K}_{\text{CSS2}}} \otimes_T \mathbb{F}_{j/B}^{\mathcal{K}_{\text{CSS2}}} \right](\mathbf{q}), \quad (25)$$

with the appropriate replacement of one or both TMDPDFs by TMD fragmentation functions for SIDIS or e^+e^- observables. Examples include

$$\text{Drell-Yan: } F_{q/A} \otimes_T F_{\bar{q}/B} + (q \leftrightarrow \bar{q}), \quad (26)$$

$$\text{SIDIS: } F_{i/A} \otimes_T D_{h/j}, \quad (27)$$

$$e^+e^- \text{ back-to-back hadrons: } D_{h_1/i} \otimes_T D_{h_2/j}, \quad (28)$$

$$\text{gluon color singlets: } F_{g/A} \otimes_T F_{g/B}. \quad (29)$$

Drell-Yan fixed-target data are therefore a useful first testbed, not a defining limitation of the formalism.

For a full transverse-momentum spectrum, the singular W term must be matched to fixed order by a nonsingular remainder,

$$d\sigma = d\sigma_W^{\mathcal{K}_{\text{CSS}2}} + d\sigma_Y, \quad d\sigma_Y = d\sigma_{\text{FO}} - \left[d\sigma_W^{\mathcal{K}_{\text{CSS}2}} \right]_{\text{FO}}. \quad (30)$$

The TMD large- k_T tail in Eq. (21) is universal matching onto collinear PDFs or FFs. The Y term in Eq. (30) is a cross-section-level fixed-order remainder. These are distinct ingredients and should not be substituted for one another. Fixed-order and matched codes such as DYTurbo, MCFM/CuTe-MCFM, SCETlib, or equivalent implementations can be used to benchmark or supply the fixed-order cross section [7, 24–26].

VII. MODEL-AGNOSTIC NONPERTURBATIVE MODULE

The perturbative prescription does not require a specific nonperturbative parametrization. At a boundary scale (μ_T, ζ_T) , one may write schematically

$$\mathbb{B}_{i/h}(x, \mathbf{k}; \mu_T, \zeta_T) = \left[\mathcal{C}_{i \leftarrow j} \otimes_x f_{j/h} \right] (x, \mathbf{k}) + \mathcal{M}_{i/h}^{\text{NP}}(x, \mathbf{k}; \theta) - \mathcal{M}_{i/h}^{\text{overlap}}(x, \mathbf{k}; \theta), \quad (31)$$

or use a convolutional smearing form,

$$\mathbb{B}_{i/h} = \left[\mathcal{C}_{i \leftarrow j} \otimes_x f_{j/h} \right] \otimes_T S_{i/h}^{\text{NP}} + \text{power-suppressed remainder}. \quad (32)$$

Other choices, including nonperturbative modifications of the Collins-Soper kernel, are possible if they are specified in the chosen scheme and do not alter the perturbative ingredient manifest.

The nonperturbative module may be a neural network, spline basis, Gaussian mixture, analytic ansatz, lattice-informed input, or another parametrization. The perturbative accuracy label belongs to the operator defined in the previous sections. It does not depend on whether the nonperturbative module is a DNN. Positivity, when imposed, should apply to physical cross sections or to well-defined low-scale model components, not to individual renormalized plus-distribution terms.

VIII. REGULATED NORMALIZATION AND MOMENTS

A naive all- k_T normalization condition,

$$\int d^2\mathbf{k} F_{i/h}(x, \mathbf{k}; \mu, \zeta) = f_{i/h}(x, \mu), \quad (33)$$

should not be imposed as an exact, unregulated identity once perturbative tails are present. The large- k_T tail produces ultraviolet-sensitive integrals and moments. Instead, one should define regulated cumulants or smeared moments,

$$\mathcal{I}_K[F_{i/h}] = \int d^2\mathbf{k} w_K(k_T) F_{i/h}(x, \mathbf{k}; \mu, \zeta), \quad (34)$$

with matching relation

$$\mathcal{I}_K[F_{i/h}] = \sum_j Z_{i \leftarrow j}(K; \mu, \zeta) \otimes_x f_{j/h}(x, \mu) + \mathcal{O}\left(\left(\frac{\Lambda_{\text{QCD}}}{K}\right)^p\right). \quad (35)$$

Similarly, transverse-momentum widths should be defined with cut or smeared weights,

$$\langle k_T^2 \rangle_K = \frac{\int d^2\mathbf{k} w_K(k_T) k_T^2 F(x, \mathbf{k})}{\int d^2\mathbf{k} w_K(k_T) F(x, \mathbf{k})}. \quad (36)$$

This prescription applies to both fixed-target and collider kinematics, with the scale K and weights chosen to maintain perturbative control.

IX. VALIDATION HIERARCHY AND CLAIM DISCIPLINE

A pure- k_T implementation should pass a validation ladder before it is used for extraction:

1. distribution algebra and bin-integral tests for $\mathcal{L}_n$ and $\mathfrak{L}_n$;
2. one-loop singular-spectrum checks in the diagonal Drell-Yan channel;
3. flavor-vector toy-PDF tests separating x -space and transverse plus prescriptions;
4. evolution path-independence and semigroup tests for Eq. (16);
5. coefficient-import tests for Eq. (23);
6. fixed-order expansion checks through the order required by the claimed accuracy;
7. comparison of binned singular spectra to the parent CSS2 implementation;

TABLE II. Claim discipline for the formalism and future implementations.

Claim	Requirement
Formal $N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ prescription	algebra, evolution, import map, manifest
Numerical $N^3\text{LL}'_{\mathcal{K}_{\text{CSS2}}}$ implementation	physical a_s^3 payload and expansion gate
$N^3\text{LL}'$ extraction	validated implementation plus data fit
DNN-based $N^3\text{LL}'$ extraction	same perturbative operator; DNN in $\mathcal{M}_{\text{NP}}$

8. full $W + Y$ matching checks against independent fixed-order or resummed benchmarks.

A compact claim table is given in Table II.

A companion validation suite was used in constructing this draft and is intended to be released as a public GitHub/Zenodo repository before submission. In the present draft, the repository citation is a placeholder and the local source bundle contains the algebraic validators. The public repository should record coefficient-file hashes, source metadata, and the expansion-gate status.

X. CONCLUSIONS

We have given a pure transverse-momentum-space prescription for CSS2-equivalent TMD factorization. The formulation replaces the local b_T -space Sudakov factor by a transverse convolution exponential acting on k_T -space distributions. Matching coefficients are imported by an analytic coefficient-level projection $L_b^m \mapsto \mathfrak{L}_{m-1}$, and the perturbative accuracy label is inherited by ingredient equivalence to the parent CSS2 calculation.

This framework separates the perturbative operator from the nonperturbative module. Consequently, a DNN can be used for the intrinsic transverse-momentum input without being part of the definition of $N^3\text{LL}'$ accuracy. The same formalism also supports analytic models, splines, Gaussian mixtures, or lattice-informed inputs. Future phenomenological work must stage the physical three-loop matching coefficients, pass the $\mathcal{O}(a_s^3)$ expansion gate, implement full $W + Y$ matching, and then perform the data extraction. The formalism developed here is designed to be the perturbative backbone for that program.

Appendix A: One-loop diagonal check

For bins away from $q_T = 0$, the one-loop diagonal Drell-Yan singular structure obtained from the k_T kernels is

$$\pi a_s [4C_F \mathcal{L}_1(\mathbf{q}; Q) - 6C_F \mathcal{L}_0(\mathbf{q}; Q)] = \frac{\alpha_s}{\pi} \frac{1}{q_T^2} \left[C_F \ln \frac{Q^2}{q_T^2} - \frac{3}{2} C_F \right], \quad (\text{A1})$$

which reproduces the standard $A_q^{(1)} = C_F$ and $B_q^{(1)} = -3C_F/2$ singular coefficients. This check fixes the signs and normalizations of the $\mathcal{L}_0, \mathcal{L}_1$ basis and the conversion between $d^2\mathbf{q}$ and dq_T^2 spectra.

Appendix B: Companion-code placeholder

The public repository entry below is intentionally a placeholder in this draft. It should be replaced by a citable archived release before journal submission.

k_T Perturbative Extraction Project, *KCSS validation suite*, GitHub repository
and Zenodo archive, URL and DOI to be supplied before submission.

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